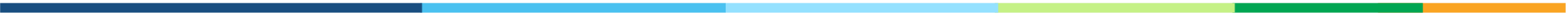




SECOND TO EIGHTH WEEK OF DEVELOPMENT

Module 1



Langman's Medical Embryology, 15th edition, pp. 51-97

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LEARNING OBJECTIVES

1. Name and briefly describe the layers of the trophoblast.
2. Describe the differentiation of the embryoblast to form the bilaminar germ disc.
3. Describe the formation of the amniotic cavity and the primitive and secondary (definitive) yolk sac.
4. Briefly describe the establishment of uteroplacental circulation.
5. Define and describe gastrulation.
6. Describe the formation of the notochord.
7. Define prechordal plate, cloacal membrane, and allantois.
8. List the derivatives of the ectoderm, mesoderm, and endoderm.
9. Describe the formation of the neural tube and neural crests.
10. Describe the differentiation of the intraembryonic mesoderm and the somites.

DEVELOPING HUMAN ORGANISM

A developing human organism during the first eight weeks of growth is called an **embryo**.

After the eight weeks, which is from the ninth week to birth, it is called a **fetus**.

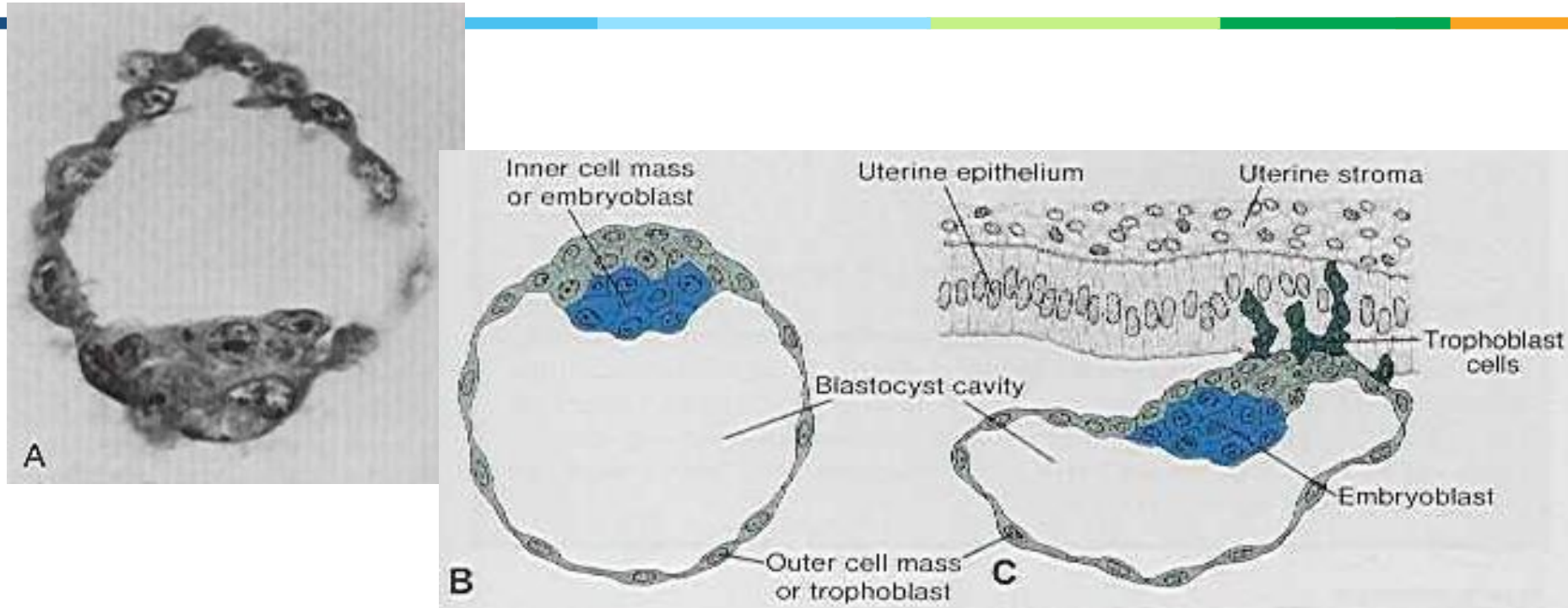
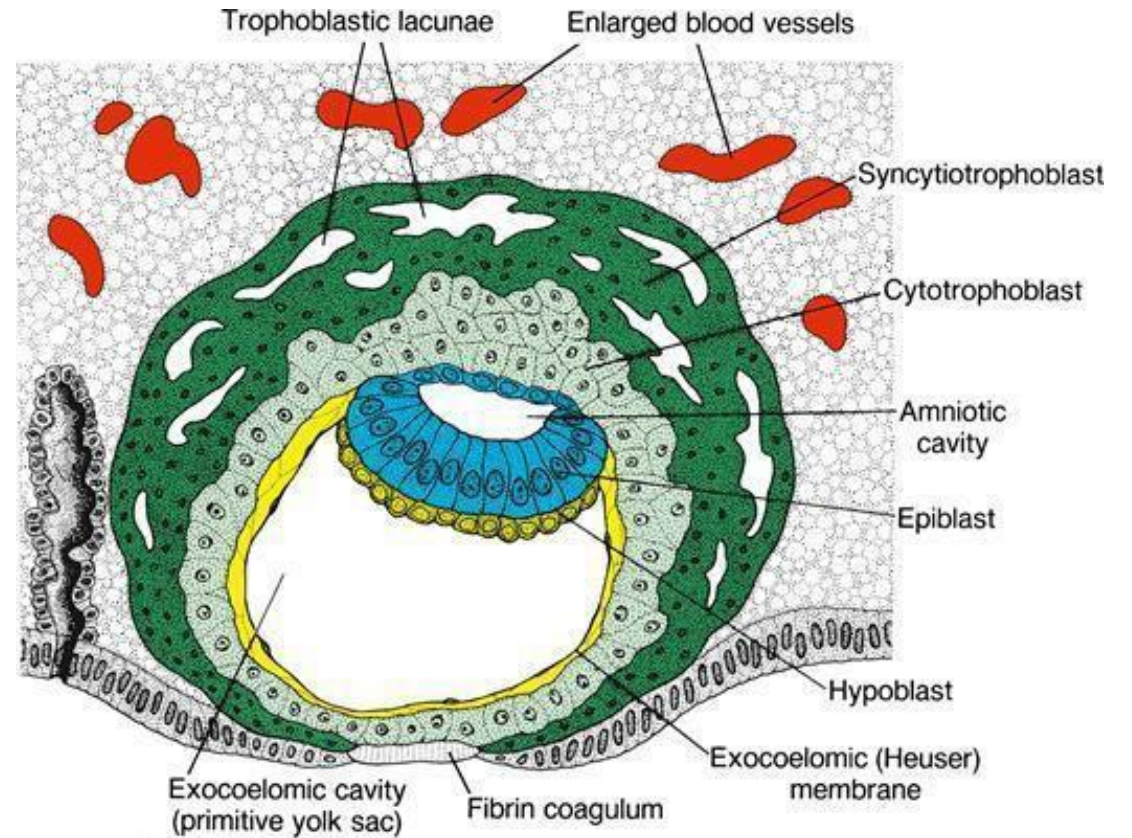


Figure 2.10. A. Section of a 107-cell human blastocyst showing inner cell mass and trophoblast cells. B. Schematic representation of a human blastocyst recovered from the uterine cavity at approximately 4.5 days. *Blue*, inner cell mass or embryoblast; *green*, trophoblast. C. Schematic representation of a blastocyst at the ninth day of development showing trophoblast cells at the embryonic pole of the blastocyst penetrating the uterine mucosa. The human blastocyst begins to penetrate the uterine mucosa by the sixth day of development.

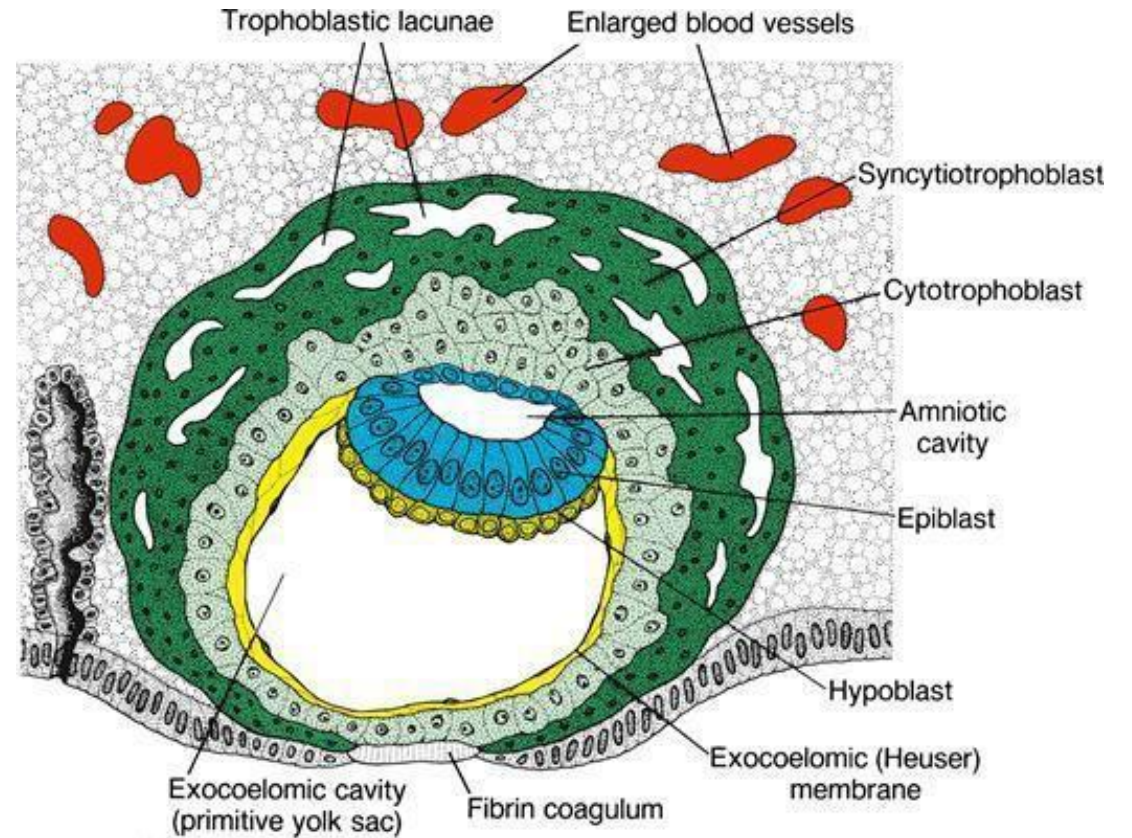
SECOND WEEK OF DEVELOPMENT

- **Formation of Bilaminar Disc;**
- Cells of the inner cell mass or embryoblast also differentiate into two layers:
- (1) a layer of small cuboidal cells adjacent to the blastocyst cavity, known as the **hypoblast layer**.
- (2) a layer of high columnar cells adjacent to the amniotic cavity, the **epiblast layer**.

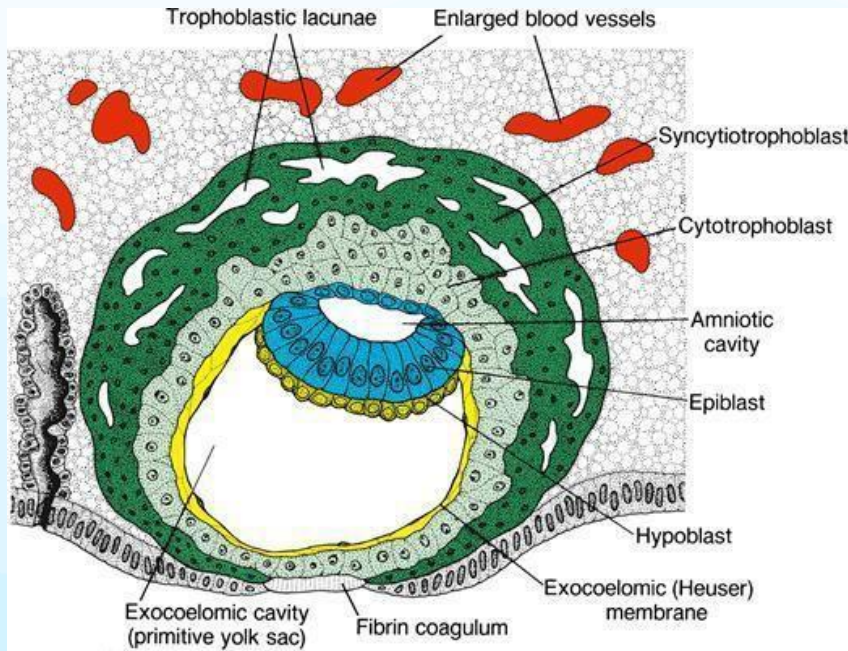


SECOND WEEK OF DEVELOPMENT

- Epiblast cells give rise to **amnioblasts** that line the **amniotic cavity** superior to the epiblast layer.



Second week of development



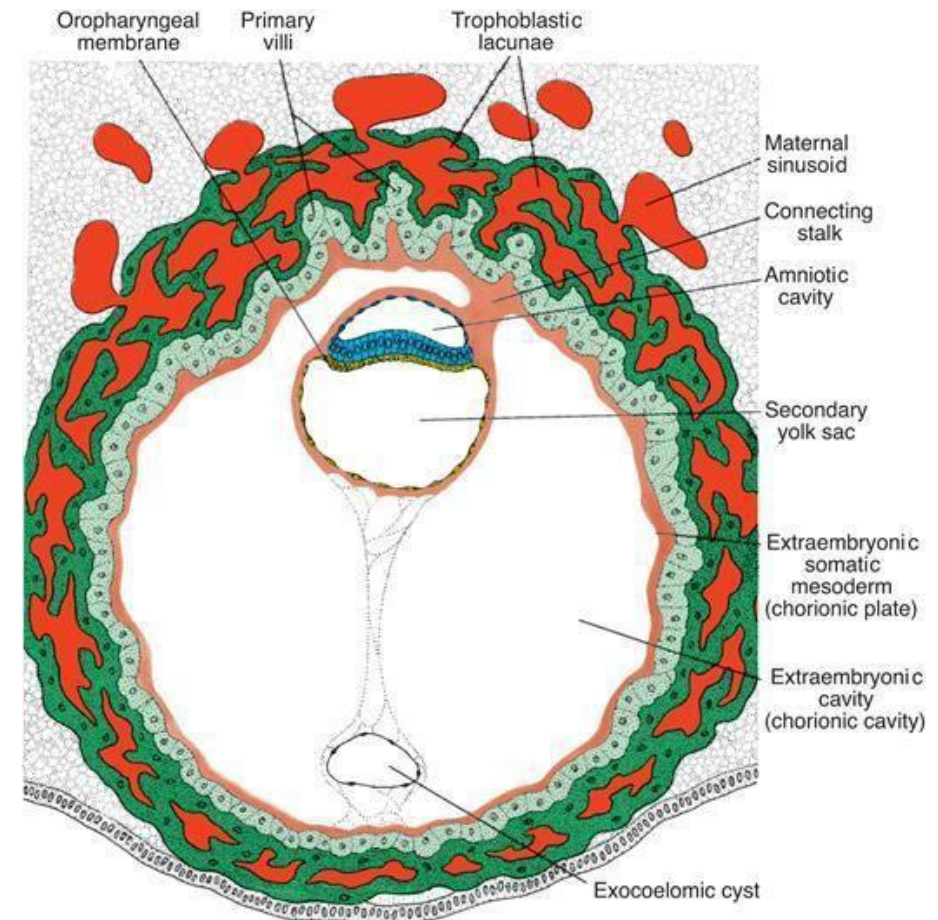
The trophoblast has differentiated into two layers:

1. an inner layer of mononucleated cells, the **cytotrophoblast**, and

2. an outer multinucleated zone without distinct cell boundaries, the **syncytiotrophoblast**.

SECOND WEEK OF DEVELOPMENT

- Lacunae develop in the syncytiotrophoblast and subsequently, maternal sinusoids are eroded by the syncytiotrophoblast. Maternal blood enters the lacunar network, and by the end of the second week, a primitive **uteroplacental circulation** begins.
- The cytotrophoblast, meanwhile, forms cellular columns penetrating into and surrounded by the syncytium. These columns are **primary villi**. By the end of the second week, the blastocyst is completely embedded, and the surface defect in the mucosa has healed

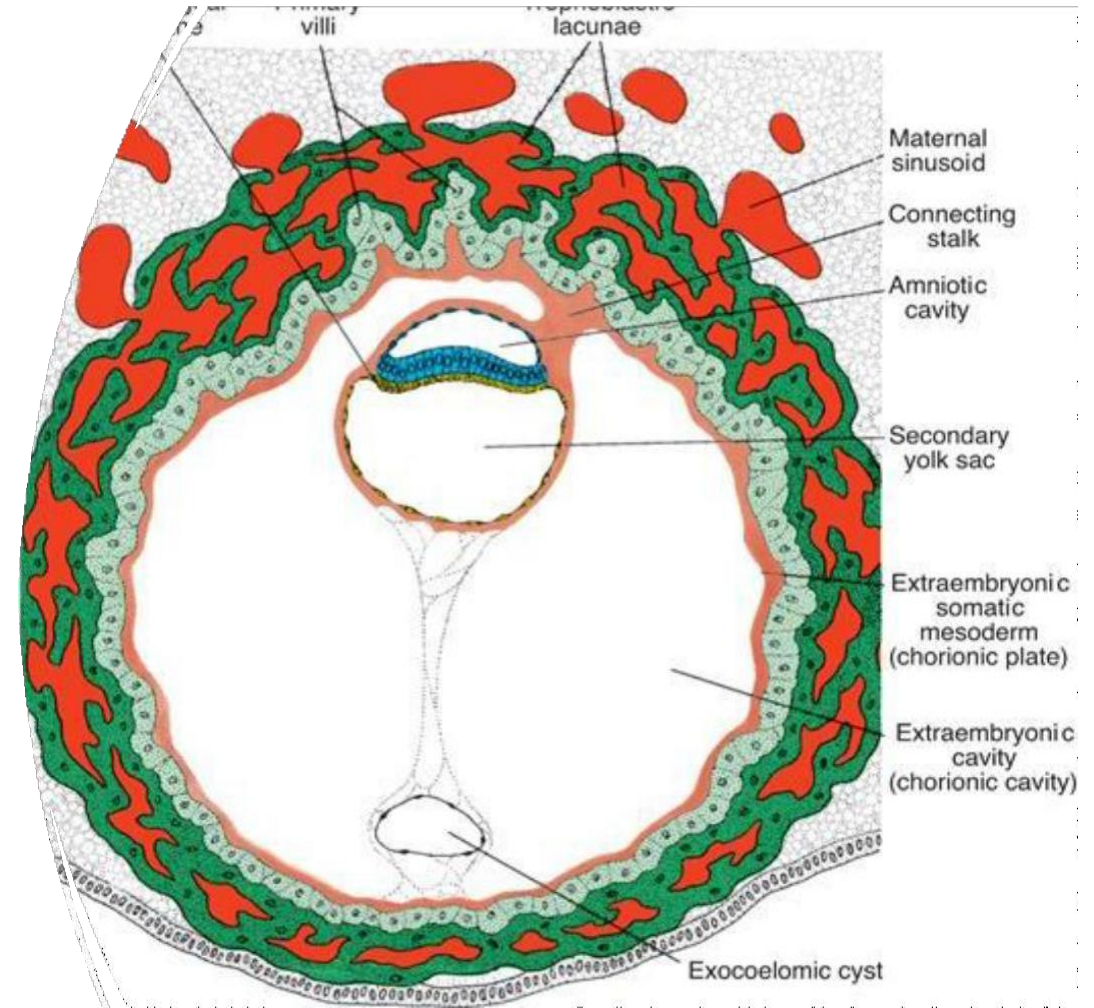


SECOND WEEK OF DEVELOPMENT

Syntiotrophoblast secretes 3 types of substances:

1. TNF-Tumor necrosis factor
2. EPF-Early pregnancy factor
3. HCG-Human Chorionic gonadotropin

Day 13 -- chorionic cavity is formed.



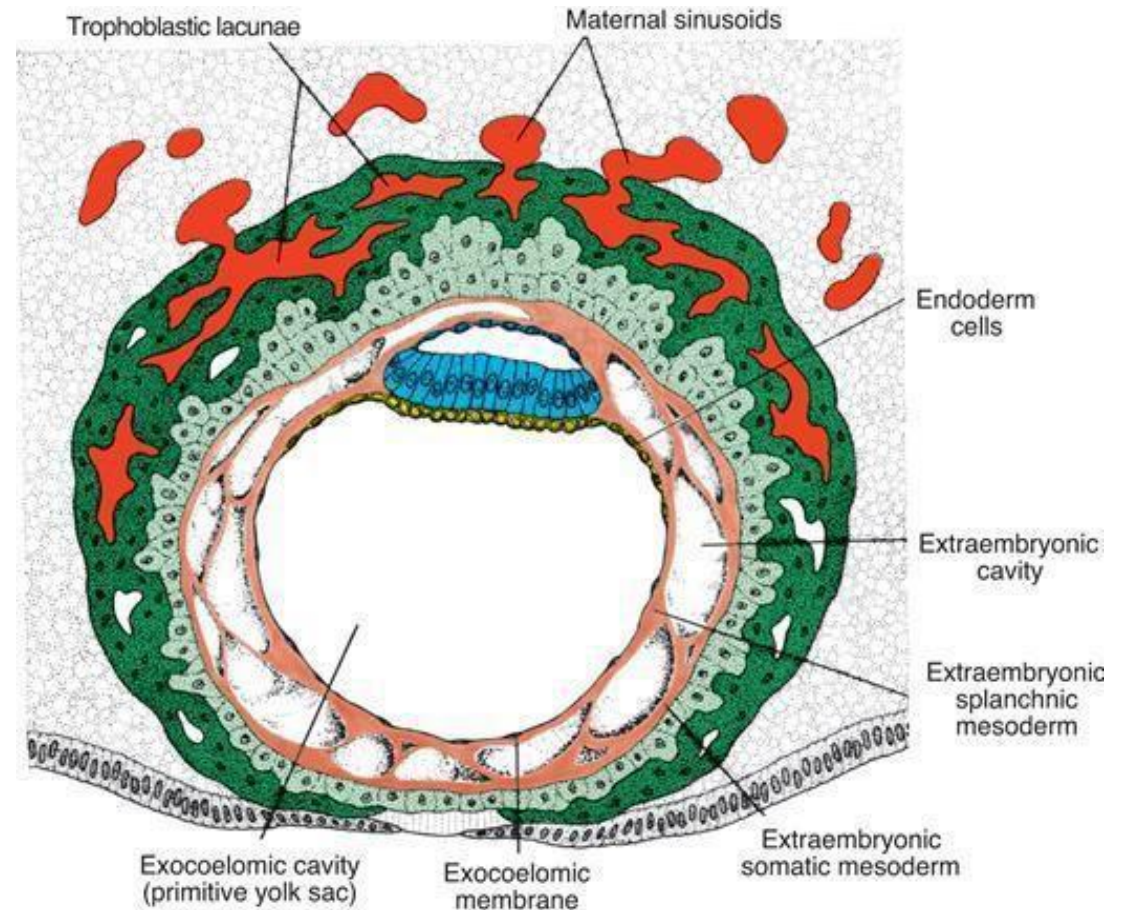
SECOND WEEK OF DEVELOPMENT

TROPHOBLAST DEVELOPMENT

By the end of the second week, the blastocyst is completely embedded, and the surface defect in the mucosa has healed.

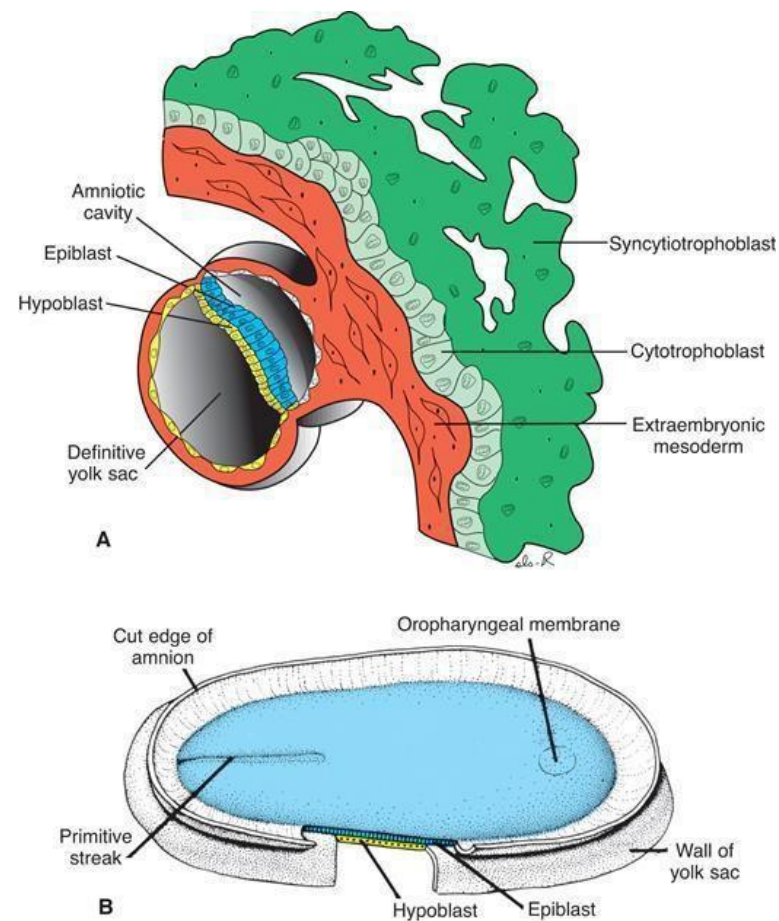
Bleeding in implantation site may occur, which may be mistaken as menstruation.

Decidual reaction – endometrial connective tissue cells swell due to accumulation of glycogen and lipid.



THIRD WEEK OF DEVELOPMENT

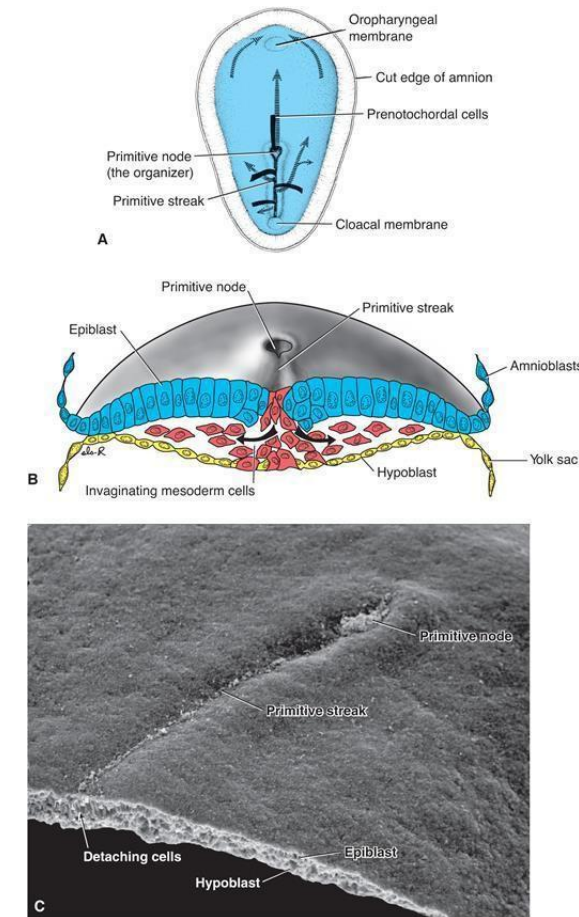
- The most characteristic event occurring during the third week of gestation is **gastrulation**, the process that establishes all three **germ layers** (**ectoderm**, **mesoderm**, and **endoderm**) in the embryo.
- Gastrulation begins with formation of the **primitive streak** on the surface of the epiblast.
- The cephalic end of the streak, the **primitive node**, consists of a slightly elevated area surrounding the small **primitive pit**



GASTRULATION

- Cells of the epiblast migrate toward the primitive streak.
- Upon arrival in the region of the streak, they become flask-shaped, detach from the epiblast, and slip beneath it.
- This inward movement is known as **invagination**.
- Once the cells have invaginated, some displace the hypoblast, creating the embryonic **endoderm**, and others come to lie between the epiblast and newly created endoderm to form **mesoderm**.
- Cells remaining in the epiblast then form **ectoderm**.

Thus, the epiblast, through the process of gastrulation, is the source of all the germ layers, and cells in these layers will give rise to all the tissues and organs in the embryo.



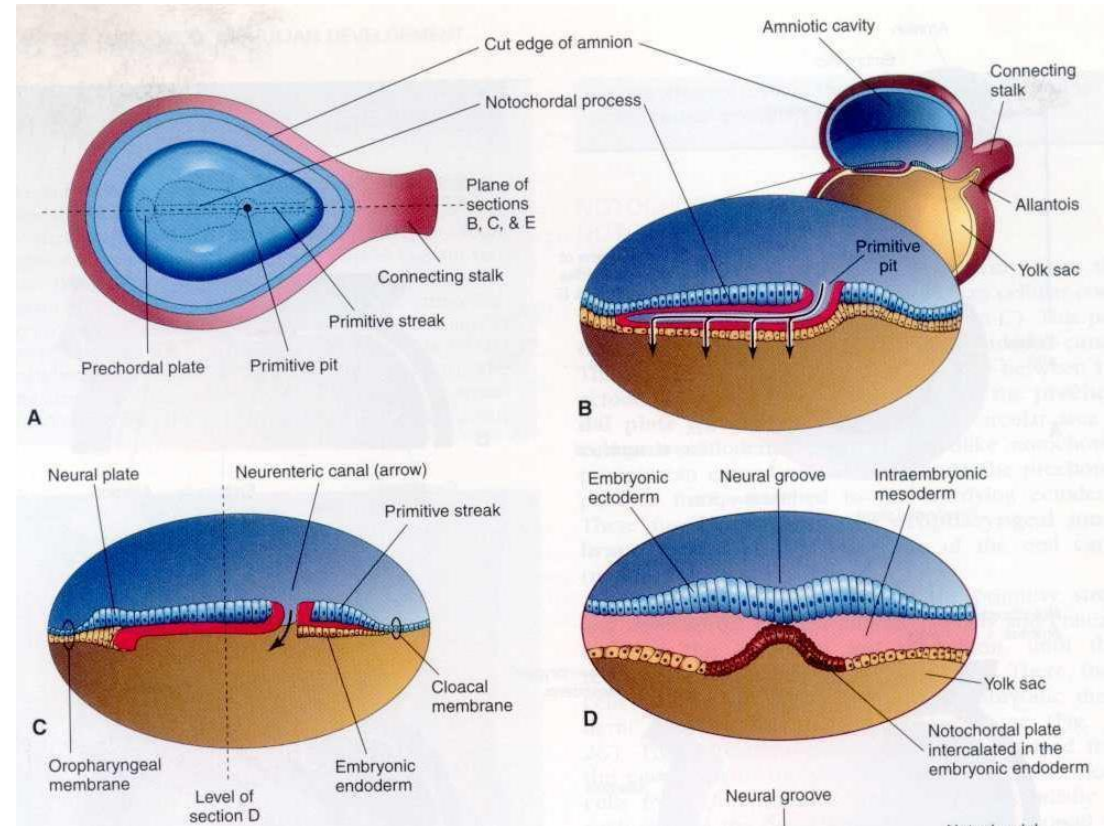
- The primitive streak degenerates at about the 4th week of development. If it persists, it can lead to a sacrococcygeal teratoma.



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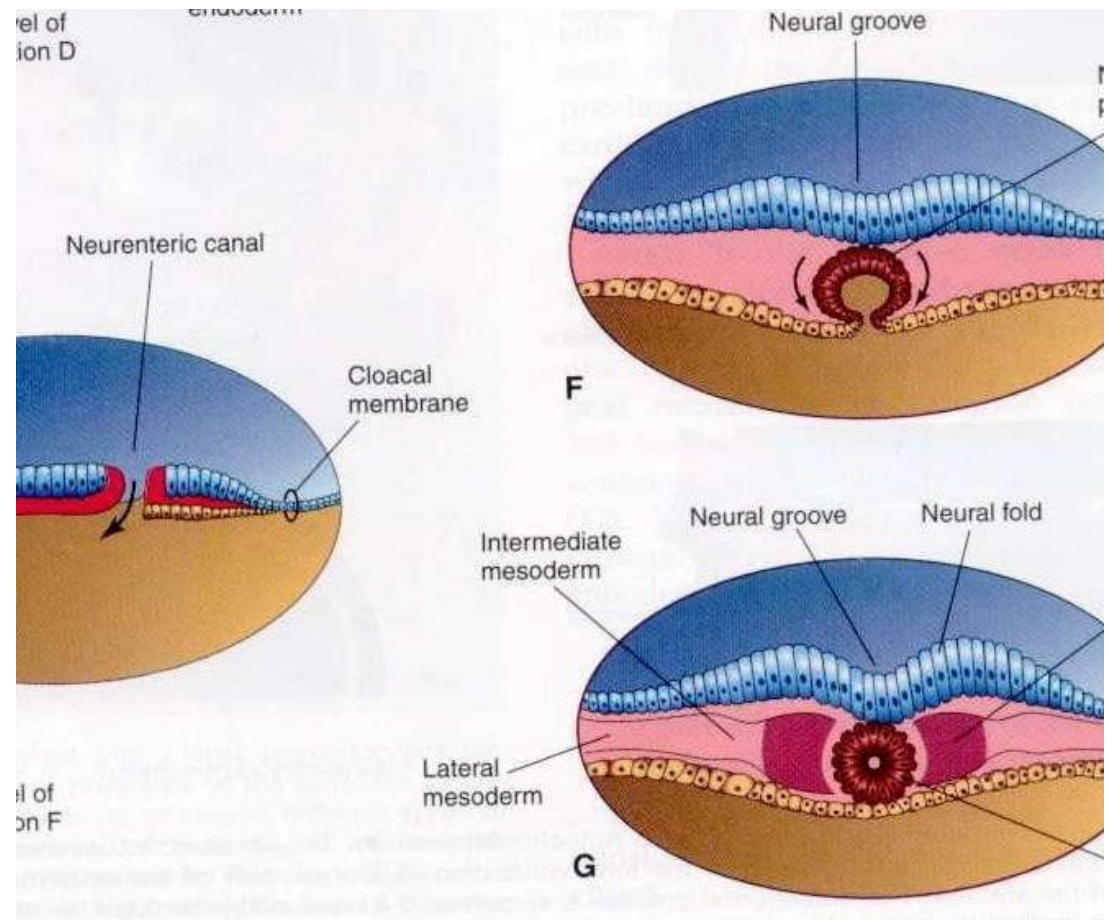
FORMATION OF THE NOTOCHORD

- About day 16, cells of primitive pit migrate rostrally toward the prechordal plate, forming a tube-like notochordal process.
- The floor of this process disintegrates leaving behind a grooved notochordal plate
- This plate invaginates craniocaudally to form notochord.

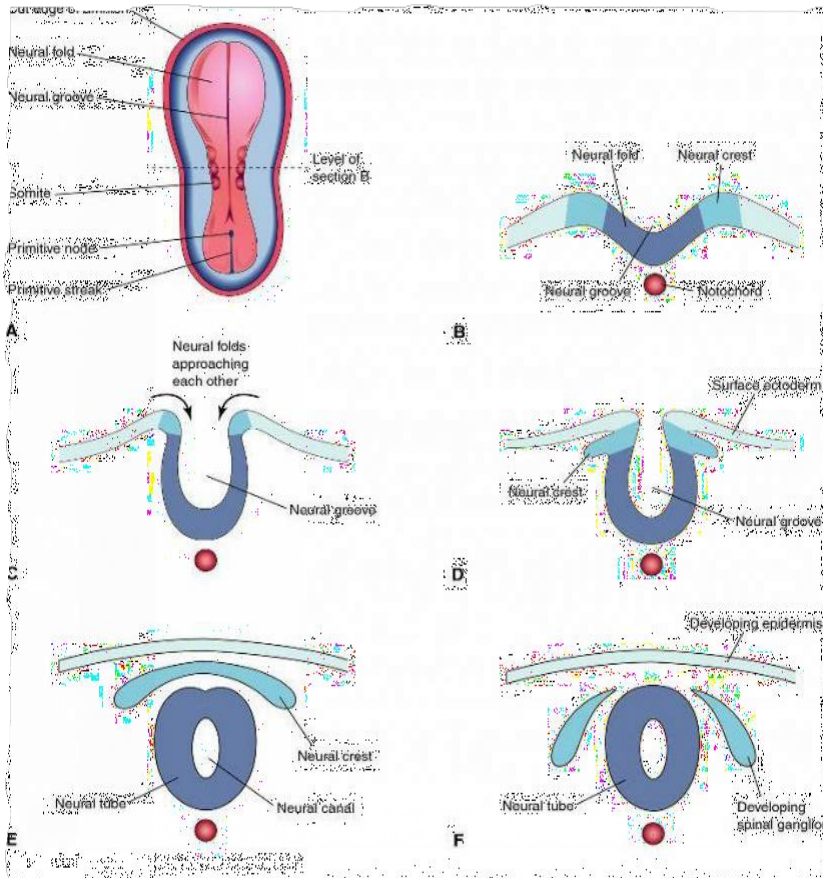


SIGNIFICANCE OF NOTOCHORD

- Provides structural rigidity to the embryo
- Forms the basis for the development of the spinal column
- Sends signaling chemicals to ectoderm to form the neural tube



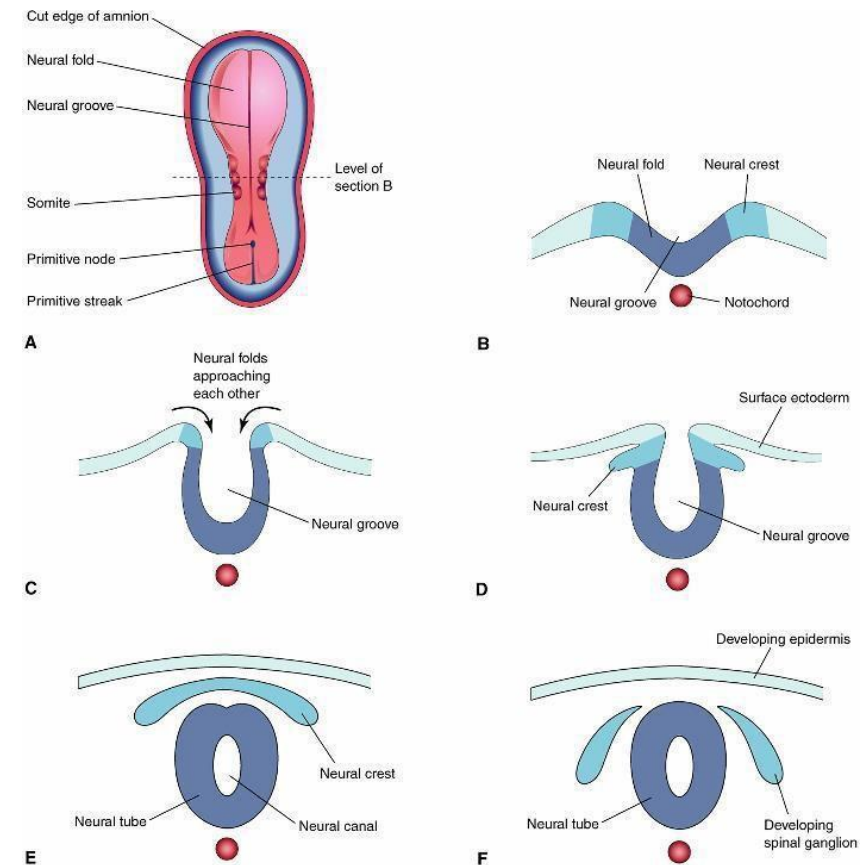
THIRD WEEK OF DEVELOPMENT



- **Neurulation:** Appearance of the notochord induces the overlying ectoderm to thicken and form the **neural plate**. Cells of the plate make up the **neuroectoderm**, and their induction is the initial event in this process.
- As the neural plate lengthens, its lateral edges elevate to form **neural folds**, and the depressed mid-region forms the **neural groove**.
- Gradually, the neural folds approach each other in the midline, where they fuse. Fusion begins in the cervical region and proceeds cranially and caudally. As a result, the **neural tube** is formed.

Neural Crest Formation:

neuroectodermal cells migrate dorsolaterally to form the neural crest between the surface ectoderm and the neural tube; it gives rise to the sensory ganglia of cranial and spinal nerves, the autonomic ganglia, sheaths of peripheral nerves, meningeal coverings of brain and spinal cord, etc.

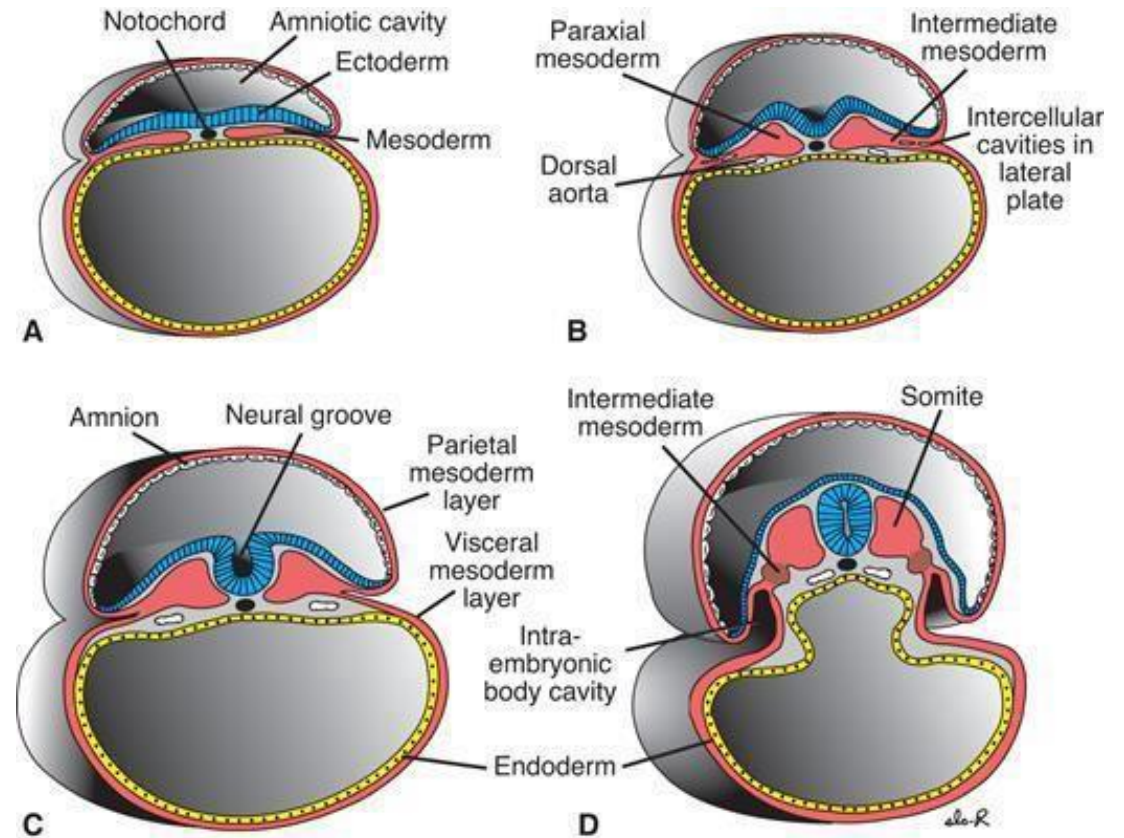


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THIRD WEEK OF DEVELOPMENT

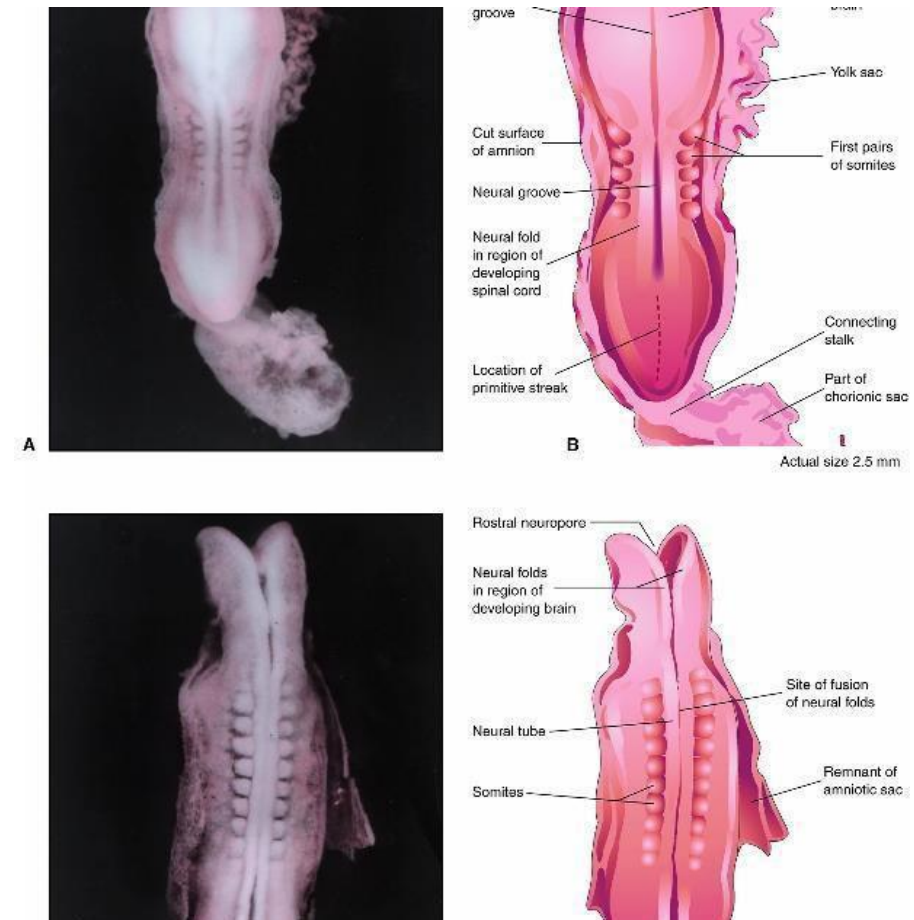
Initially, cells of the mesodermal germ layer form a thin sheet of loosely woven tissue on each side of the midline. By approximately the 17th day, however, cells close to the midline proliferate and form a thickened plate of tissue known as **paraxial mesoderm**. More laterally, the mesoderm layer remains thin and is known as the **lateral plate**.

- A layer continuous with mesoderm covering the amnion, known as the **somatic** or **parietal mesoderm layer**
- A layer continuous with mesoderm covering the yolk sac, known as the **splanchnic** or **visceral mesoderm layer**.
- **Intermediate mesoderm** connects paraxial and lateral plate mesoderm.



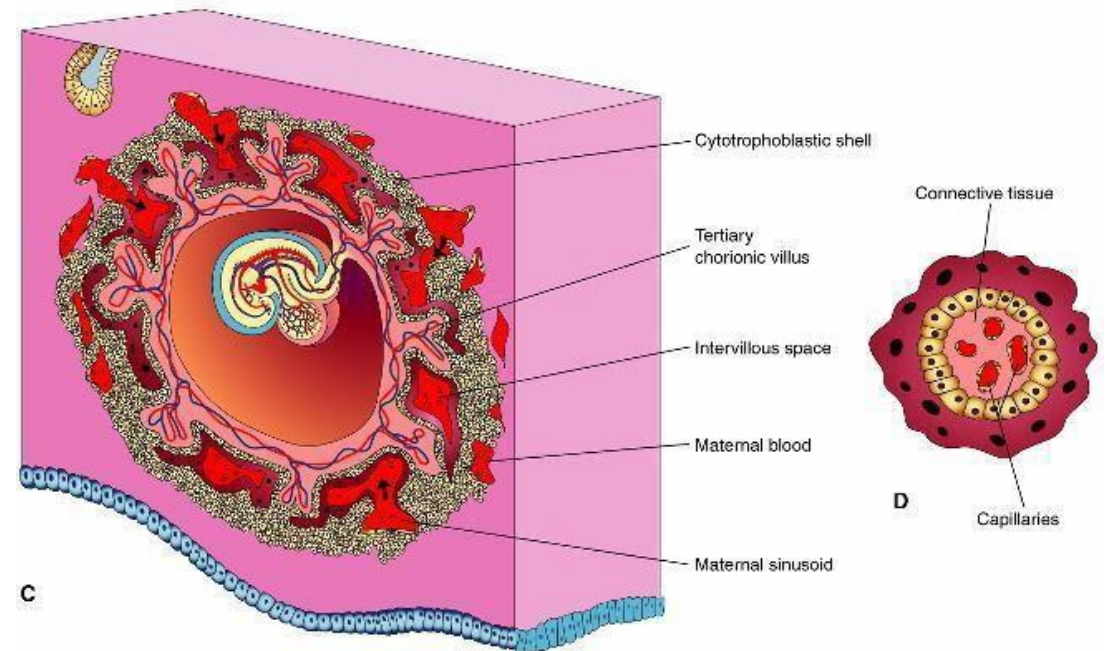
SOMITE FORMATION

- From the occipital region caudally, paraxial mesoderm becomes segmented and organizes into **somites**. The first pair of somites arises in the occipital region of the embryo at approximately the 20th day of development. From here, new somites appear in craniocaudal sequence at a rate of approximately three pairs per day until, at the end of the fifth week, 42 to 44 pairs are present.
- Somites give rise to the **myotome** (muscle tissue), **sclerotome** (cartilage and bone), and **dermatome** (dermis of the skin), which are **all supporting tissues of the body**.

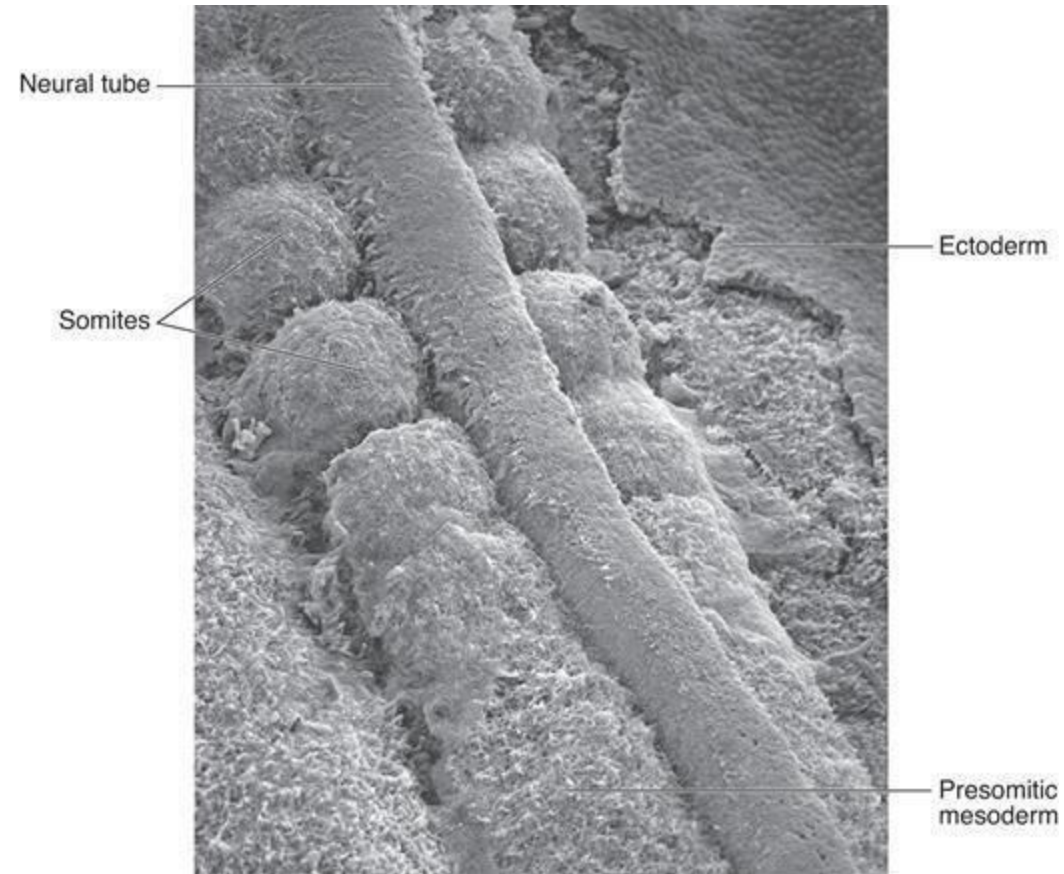


THIRD WEEK OF DEVELOPMENT

- . TROPHOBLAST DEVELOPMENT
- **the extraembryonic mesoderm of the chorion invades the cytotrophoblastic core of the primary villi --> secondary villi.** The core mesoderm of these villi organize into capillaries and make contact with the connecting stalk and chorion --> tertiary villi.

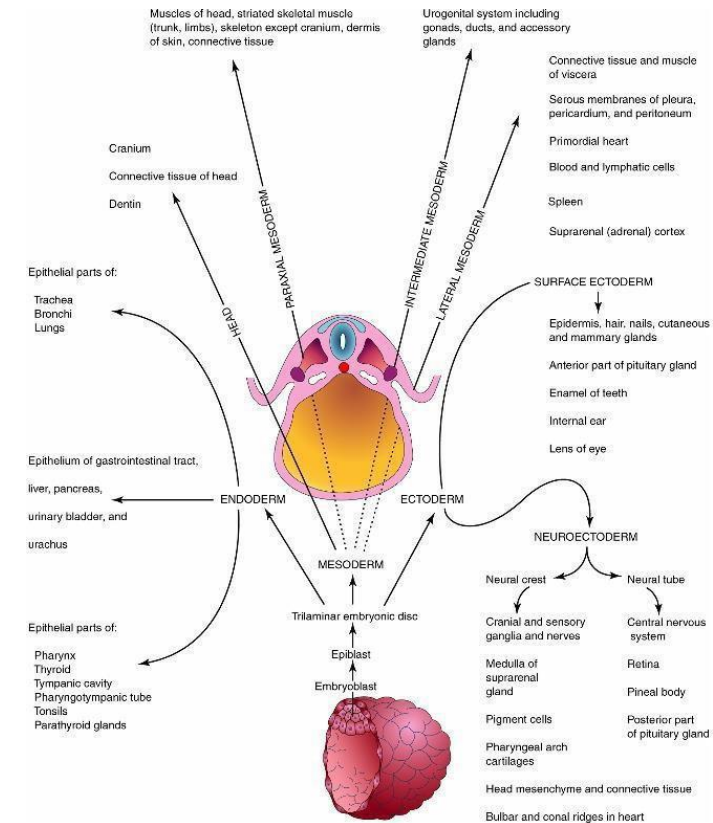


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EMBRYONIC PERIOD

During the **3rd to 8th week of development**, the embryo undergoes a period of **organogenesis**, in which each of the three germ layers gives rise to specific tissues and organs.



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ORGANOGENESIS

DERIVATIVES OF ECTODERM:

- Epidermis of skin and skin appendages.
- All nervous tissue.
- Lens, cornea, and intraocular muscles.
- Internal and external ear.
- Neuroepithelium of sense organs.
- Epithelium of oral and nasal cavities, paranasal sinuses, salivary glands, anal canal.
- Epithelium of pineal gland, pituitary gland, and adrenal medulla.

ORGANOGENESIS

DERIVATIVES OF MESODERM:

- Dermis of skin.
- All skeletal, most smooth, and cardiac muscles.
- Cartilage, bone and other connective tissues.
- Blood, bone marrow, and lymphoid tissue.
- Endothelium of blood and lymphatic vessels.
- Fibrous and vascular layer of eye.
- Middle ear.
- Mesothelium of ventral body cavities.
- Epithelium of kidneys, ureters, adrenal cortex, gonads.

ORGANOGENESIS-MESODERM DERIVATIVES

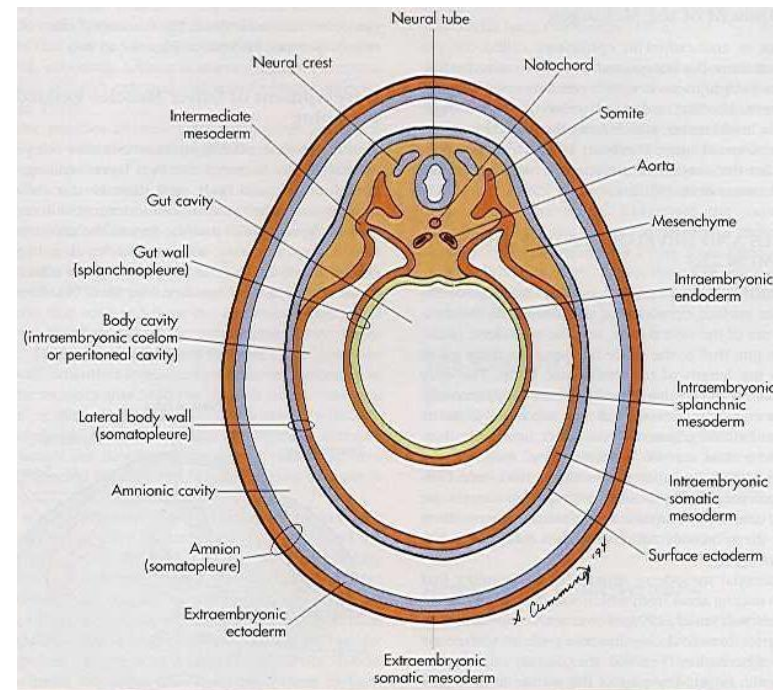
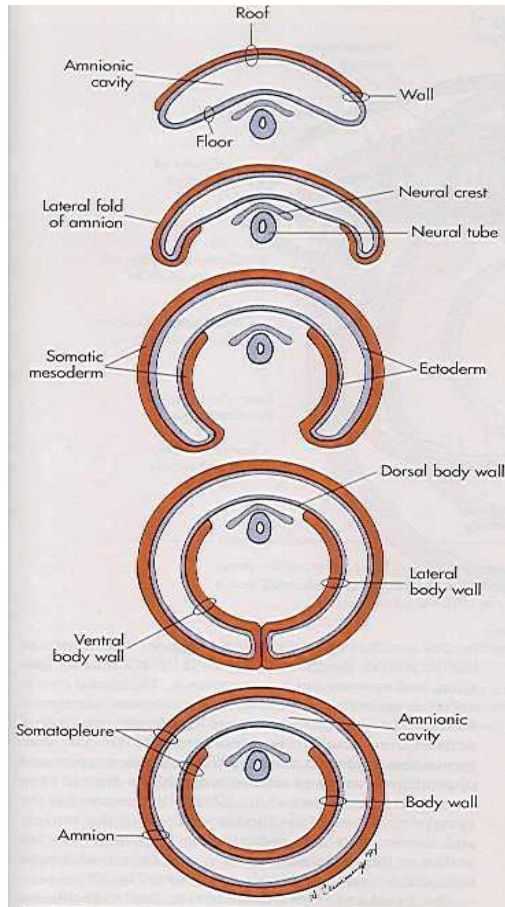
- 17th day - mesoderm is arranged into the paraxial (A), intermediate (B), and lateral plates - somatic portion & visceral portion.
- Beginning of 3rd wk. - somites begin to appear in the cephalic region and proceed caudally.
- Beginning of 4th wk. - somites differentiate into sclerotome, myotome, dermatome.
- Cervical and upper thoracic regions of the intermediate mesoderm will form nephrotomes.
- The remaining caudal portion is an unsegmented mass called the nephrogenic cord.

ORGANOGENESIS

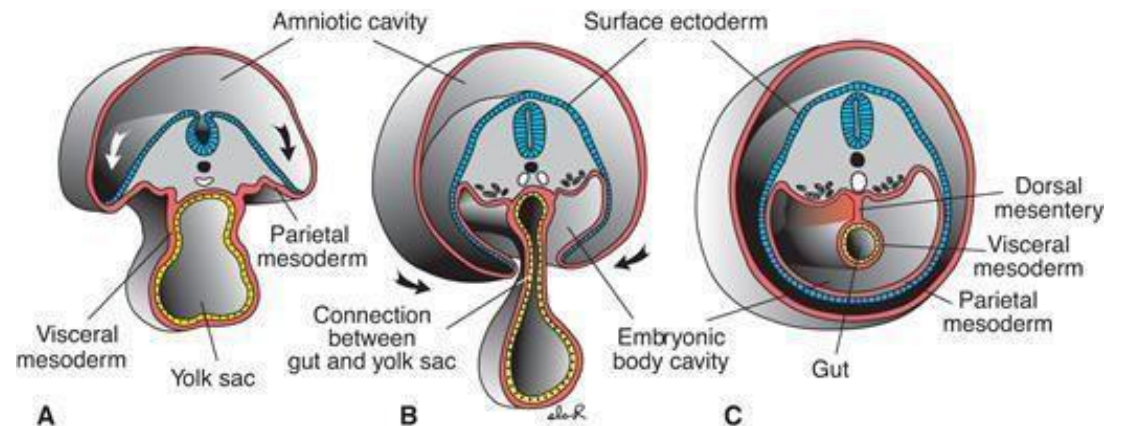
DERIVATIVES OF ENDODERM:

- Epithelium of GI tract except oral cavity & anal canal.
- Epithelium of urinary bladder, liver, & gallbladder.
- Epithelium of pharynx, auditory tubes, tonsils, larynx, trachea, bronchi, & lungs.
- Epithelium of thyroid, parathyroid, pancreas, and thymus glands.
- Epithelium of accessory glands of reproductive system.

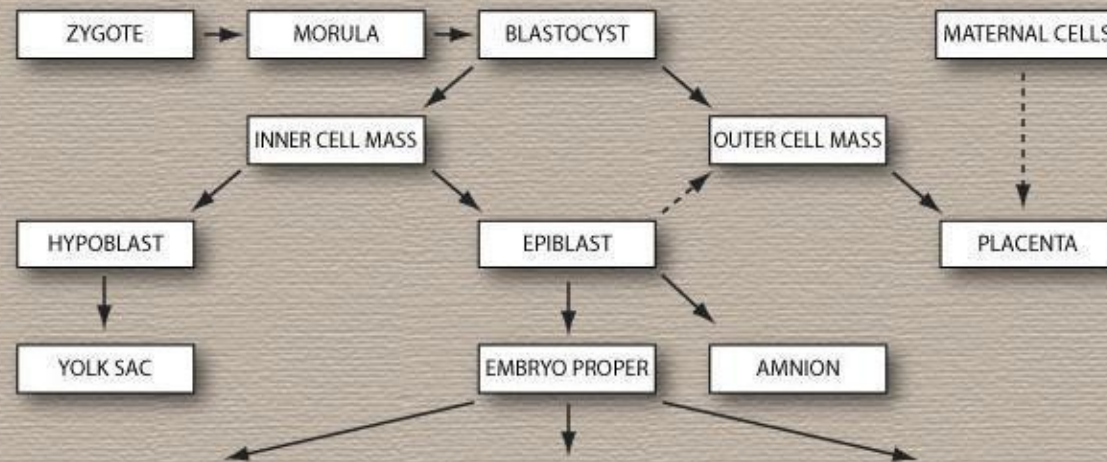
FORMATION OF PRIMITIVE GUT



- With cephalocaudal and lateral folding of the embryo, the primitive gut is formed by the incorporation of part of the endoderm-lined yolk sac.
- The primitive gut is divided into: foregut, midgut, and hindgut.
- The gut is lined externally by lateral plate mesoderm - splanchnic portion.



ORGAN AND BODY SYSTEM FORMATION



GERM LAYER DERIVATIVES		
ECTODERM	MESODERM	ENDODERM
brain	heart	lining of the respiratory system
spinal cord	kidneys	lining of the digestive tract
nerves	bones	liver
skin	cartilage	pancreas
nails	muscles	bladder
hair	blood cells	
	blood vessels	

References

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2. Before we are born: Essentials of embryology and birth defects by Moore, K. L., Persaud, T. V. N., & Torchia, M. G., 9th ed., 2016, Elsevier